

temperature goes above the upper limit, the output of IC1(B) at pin 7 goes low and LED2 goes off. This means as the temperature rises beyond the upper limit, the voltage at pins 2 and 6 falls. When the voltage at pin 6 is less than the voltage at pin 5, the output becomes high and LED2 glows.

The LED status for different water temperatures is summarised in the table. The ideal water temperature to bathe the baby is when both the LEDs glow.

The circuit requires 9V for operation, which you can get either from six AA-size or AAA-size cells in a battery holder or from a 9V PP3 battery.

Assemble the circuit on a general-

LED Status for Different Water Temperatures

Temperature	LED1	LED2
$\leq 25^{\circ}\text{C}$	Off	On
$> 25^{\circ}\text{C} \leq 40^{\circ}\text{C}$	On	On
$> 40^{\circ}\text{C}$	On	Off

purpose PCB and enclose in a suitable cabinet. The circuit board is small enough to fit inside the miniature plastic enclosure of a handheld gadget. If you want to build it as a probe-type instrument, drill a hole at one end of the enclosure and glue a short length of stiff plastic tubing (fish-tank aerating tube or a scrap of model-maker's

polystyrene tube). Extend the thermistor out from the circuit using two insulated wires. Glue the thermistor to the end of the tube. Apply plenty of glue to the thermistor so that it dries to form a waterproof seal.

For low-temperature setting, put the thermistor in a glass of tepid water. Adjust VR1 until LED1 goes off. For high-temperature setting, put the thermistor in a glass filled with hot water. Adjust VR2 until LED2 goes off. Test your settings by immersing the probe in water within the acceptable temperature range. In this case both the LEDs should glow. Fig. 2 shows the author's prototype. ●

Versatile DC-DC Converter

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Here is a versatile power coupler that connects a device to 5V-19V DC generated from AC mains by a power adaptor. Power adaptors come in different voltage outputs like 5V (for mobile phones), 12V (for external hard drives) and 19V (for laptops). Sometimes the power adaptor may have a voltage rating higher than the required voltage. With the converter circuit given here, the adaptor can be used to power any device at a lower voltage.

For instance, by using a 19V laptop adaptor, you can power a TTL circuit at 5V. There can also be other instances when one needs a 3V or 6V supply. All these and many other intermediate voltages are easily possible with this versatile converter circuit when used together with any off-hand power adaptor.

Fig. 1 shows the circuit of the DC-DC converter. Smooth reduction in the voltage is achieved using the LM317 regulator IC. The complete unit can fit inside a piece of a glue stick tube.

Adjusting variable resistor VR1 gives the desired output voltage. The output voltage is read using a 0-100 μA ammeter, whose series resistance R^* is chosen such that the maximum desired

voltage could be covered. For instance, if full-scale deflection (FSD) current of the meter is 100 μA and you need an output voltage of up to 15V, then $R^* = 15/0.0001 = 150 \text{ k}\Omega$. The desired value of R^* is obtained by using 150-kilo-ohm preset VR2.



Use of a variable resistor which also has an on/off switch like the one in old radios is recommended. It will cut off the coupler from the input power supply without having to ac-

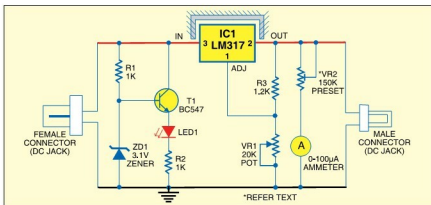


Fig. 1: Circuit of the DC-DC converter

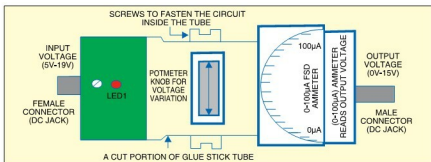


Fig. 2: Proposed assembly